INTERNSHIP PROJECT ASSIGNMENT

PYTHON BACKEND DEVELOPMENT | EWB EDU TECH PVT. LTD.

Project: Students Database CRUD Application (+ Optional AI Extension)

Project Type: Individual backend development project

01 PROJECT OBJECTIVE

Build a Python backend application to manage student records using Create, Read, Update and Delete (CRUD) operations. Use FastAPI for the API and SQLite for persistent storage. An optional Ollama extension can answer natural-language questions about the stored data.

02 MANDATORY CORE REQUIREMENTS

- Create: Add new student records to the database.
- Read: List all students and retrieve or filter individual profiles.
- Update: Modify an existing student's details.
- Delete: Remove student records from the database.
- Technology: Python, FastAPI and SQLite are required.

03 STUDENT RECORD FIELDS

Include a unique student ID, name, date of birth and contact information (email and/or phone). Optional fields include course, address and enrollment date.

04 SUGGESTED API ENDPOINTS

METHOD	ENDPOINT	PURPOSE
POST	/students	Create a student
GET	/students	List / filter students
GET	/students/{student_id}	View a student
PUT / PATCH	/students/{student_id}	Update details
DELETE	/students/{student_id}	Delete a student

These routes are examples; equivalent documented endpoints are acceptable.

05 OPTIONAL BONUS: LOCAL AI WITH OLLAMA

This section is entirely optional. The core CRUD application is sufficient to complete the assignment.

- Run a local Ollama model, for example llama3.2:1b.
- Accept questions such as “How many students are there?” or “List students born after 2005.”
- Fetch relevant database records in backend code and provide them as context to the model.
- Return an accurate, human-readable response grounded in those records.

Security: Do not let the model execute unrestricted SQL. Use dummy data and do not publish real student personal information.

06 SUBMISSION GUIDELINES

- Upload the complete project source code to a GitHub repository.
- Add a clear README.md with prerequisites, installation, run commands and API testing instructions.
- Include dependencies (e.g., requirements.txt) and sample requests or screenshots if available.
- Do not commit passwords, secrets or real students’ personal information.
- Submit the GitHub repository link using the channel announced by your EWB trainer.

07 PROJECT REVIEW CHECKLIST

- All four CRUD operations work correctly.
- SQLite data persists and can be retrieved accurately.
- Input validation and missing-record handling work properly.
- Code is organized and README instructions are reproducible.
- Optional: AI answers are grounded in retrieved records.

08 NOTES FOR STUDENTS

Prioritize a fully functional FastAPI + SQLite application. Be ready to demonstrate the API, explain your architecture and describe how records are created, updated and deleted. The AI extension is an extra challenge, not a mandatory requirement.

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